

A Shared autoencoder

3,000 HVGs -> encoder [256,128]

10-D latent representation

reconstruction objective shared

across all three variants

B Adaptive DPMM prior

90% reconstruction warmup

Bayesian mixture refit

every 10 epochs

cluster count inferred online

C Conditional flow refinement

transport toward DPMM centers

flow weight $\lambda_{FM} = 0.1$

noise scale = 0.5

projection-fidelity hypothesis

D Pre-specified operating points

Pure-AE: label concordance

DPMM-Base: compact geometry

DPMM-FM: projection fidelity

No universal-best assumption